

PROJECT PRESENTATION

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Project Title: Crop Yield Prediction Using Lasso, Ridge and Elastic Net Regression

INTRODUCTION

Overview: Crop yield is predicted from 19,689 records across 30 Indian states (1997–2020) by joining crop, soil and weather data and fitting Ridge, Lasso and Elastic Net regression.

Problem: Yield depends on interacting factors such as rainfall, temperature, soil and inputs. Can regularised linear models predict it when the inputs are highly correlated?

Objectives: Merge three datasets, build 13 input features, then train and compare three models on MAE, MSE, RMSE and R^2.

Scope: State-level soil and weather data, tested on unseen years (2016–2020). Supports planning, not field-level agronomic advice.

RESEARCH GAP

Limitation: Simple averages and rules miss multi-variable effects. Area, fertilizer and pesticide correlate at $r \geq 0.95$, yet each correlates below 0.07 with yield.

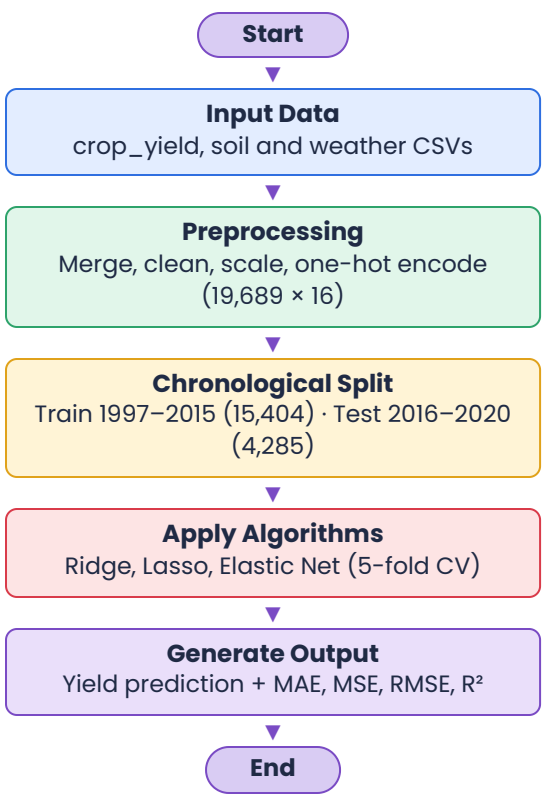
Unexplored: L1, L2 and L1+L2 penalties compared side by side on joined crop, soil and weather data.

Need: Leakage-free evaluation: production is excluded and the split is chronological, not random.

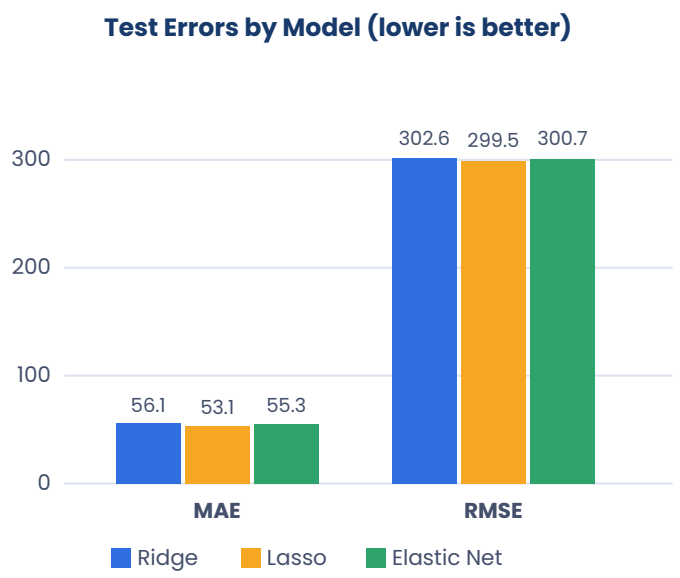
Our work: Lasso is marginally best (R^2 0.876, RMSE 299.5) and zeroes 56 of 100 encoded features. The three models differ only slightly.

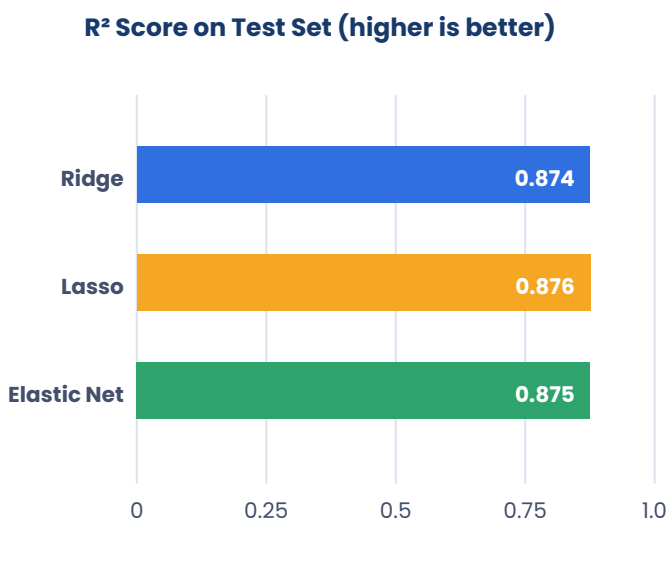
19,689	30	13	0.876
Records merged (1997–2020)	Indian states covered	Input features (10 numeric + 3 categorical)	Best R^2 (Lasso, test set)

ALGORITHM



GRAPHS

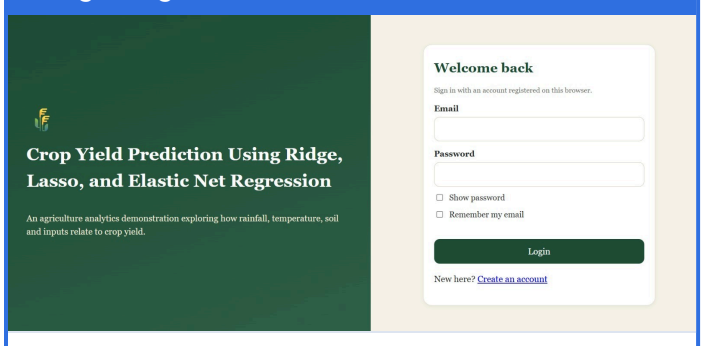




Chronological test set, 2016–2020 (4,285 records). Lasso has the lowest errors and highest R^2, but gains over Ridge and Elastic Net are small (R^2 within 0.002).

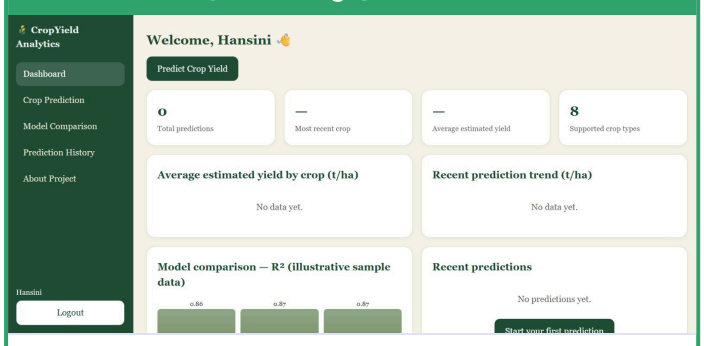
RESULTS – FRONT END

1. Login Page



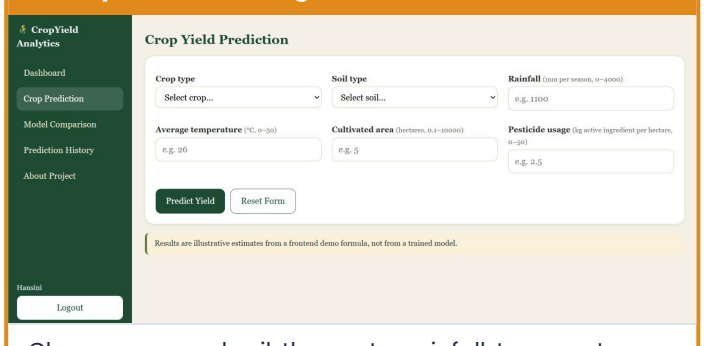
Email and password sign-in, with show-password, remember-email and create-account options (accounts are stored in the browser).

2. Dashboard (Home Page)



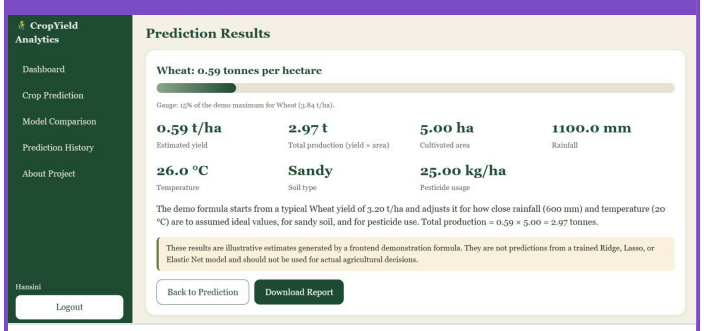
Summary cards (predictions, recent crop, average yield, 8 supported crops), yield charts and a model R^2 comparison.

3. Crop Prediction Page



Choose crop and soil, then enter rainfall, temperature, cultivated area and pesticide usage. Predict Yield or Reset Form.

4. Prediction Results



Estimated yield in t/ha, total production (yield × area), a yield gauge and a Download Report button.

Key Front-End Features

- ✔ Login and create-account screens
- ✔ Dashboard with prediction count, recent crop and average yield
- ✔ Prediction form with input ranges: rainfall 0–4000 mm, temperature 0–50 °C, area 0.1–10,000 ha, pesticide 0–50 kg/ha
- ✔ Results with yield per hectare, total production and gauge
- ✔ Downloadable prediction report
- ✔ Pages for Model Comparison, Prediction History and About Project

> The front end shows illustrative estimates from a demo formula, not from the trained Ridge, Lasso or Elastic Net models. The reported metrics come from the trained models above.